UCR RESEARCH COMPUTING · QUICK START

Run your research on Nautilus, from your AI assistant

Connect your laptop to the National Research Platform, give your assistant the **nrp** tools, then move a Slurm job to Kubernetes by asking in plain language.

nrp-mcp 1.1.0 or newer
macOS · Linux · Windows
KB031 · Oct 2026

National GPUs and CPUs
No UCR recharge and no allocation proposal to write.

Plain language
Your assistant plans, runs, watches and cleans up for you.

Nothing hidden
Every plan shows what it creates and the kubectl it equals.

Your yes first
GPU, public and delete steps wait for a code only you can see.

1 Sign in to Nautilus — 2 min

Go to **nrp.ai**, click **Login**, choose **University of California, Riverside**, and accept the policy. Your first sign-in is what activates your namespace invite.

✓ **You see your name on nrp.ai; your namespace admin sees you.**

2 Install nrp-mcp — 2 min

MAC / LINUX

```
$ curl -fsSL https://raw.githubusercontent.com/UCR-Research-Computing/nrp-mcp/main/scripts/install.sh | sh
```

WINDOWS

```
PS> irm https://raw.githubusercontent.com/UCR-Research-Computing/nrp-mcp/main/scripts/install.ps1 | iex
```

Both verify the download and put `nrp-mcp` on your PATH (Mac asks: answer **y**); then open a new terminal.

✓ `nrp-mcp version` **prints 1.1.0 or newer.**

3 Connect your laptop and saved results — 8 min

Download your config at **nrp.ai/config** (it lands in Downloads), then run:

```
$ nrp-mcp setup
```

Installs kubectl and the sign-in plugin (no admin rights, checksums verified), puts the config in place, opens the UCR sign-in. Asks first.

Then save your S3 keys from **nrp.ai/s3token** so job results come home, and check it all:

```
$ nrp-mcp s3 setup
$ nrp-mcp doctor
```

✓ **Every** `doctor` **line says ok; any other line names the next step.**

4 Install Hermes Agent and add your key — RECOMMENDED — 8 min

MAC / LINUX

```
$ curl -fsSL https://hermes-agent.nousresearch.com/install.sh | bash
```

WINDOWS

```
PS> iex (irm https://hermes-agent.nousresearch.com/install.ps1)
```

Open a new terminal. Store your Gemini API key and pick the model:

```
$ hermes config set GEMINI_API_KEY your-key
$ hermes config set model.provider gemini
$ hermes config set model.default gemini-3.8-flash
```

✓ `hermes chat -q "hello"` **answers.**

5 Give Hermes the nrp tools — 2 min

```
$ nrp-mcp client hermes
```

Adds nrp (full path) to Hermes's settings, with a backup. Other apps: see the back. Then start `hermes` and ask:

"Where do I stand on Nautilus?"

✓ **It answers with your user, namespace, what is running and your GPU quota.**

Your key is yours. Never paste it into a chat, a repo, a shared doc or a screenshot.

Turn over: the exercise, what to say, the rules

HANDS-ON · SLURM TO KUBERNETES

Move a Slurm array job to Nautilus in two sentences

The exercise

Get `examples/slurm-array-bootstrap` from the project (QR below, Code, Download ZIP): an R bootstrap of a trimmed mean and a normal Slurm script with `--array=0-9`. In that folder, ask Hermes:

1. *"Plan the Slurm script in this folder and show me what each line becomes."*
2. *"Run it."* (no S3 keys yet? add "yes, discard results")
3. *"How did it go?"*
4. *"Clean up everything I made."*

SLURM LINE	BECOMES ON NAUTILUS
`--cpus-per-task=1` `--mem=2G`	CPU and memory request; limits equal requests
`--time=00:20:00`	a time limit on the Job (whole hours)
`--array=0-9`	an Indexed Job: 10 tasks, each with its own `$SLURM_ARRAY_TASK_ID`
`module load R`	a container image with R in it

Ten tasks, ten seeds, ten different answers, in about a minute. The same script still runs on Slurm.

What to say

YOU SAY	NRP DOES
"Walk me through it"	one step at a time, from wherever you are
"Where do I stand?"	who you are, what runs, quotas, warnings
"Run train.py on a GPU"	plans it, waits for your approval code, then runs
"Run this 50 times, seeds 1 to 50"	a sweep (Indexed Job)
"Why did it fail?"	logs plus a plain diagnosis and the fix
"Give me a Jupyter notebook"	a private session on Nautilus
"Copy results to my laptop"	downloads the run's saved results
"Clean up"	lists what you made, deletes after your code

Rules of the road

- **P1 data only.** Public, non-sensitive data. No student records, PII, clinical or controlled data.
- **No idle GPUs.** Ask for a GPU only when the code uses one; idle GPUs get accounts suspended.
- **Public web app? Ask your namespace admin first** (RC in a workshop namespace). It needs your approval code and expires in 14 days.
- **Clean up** when you finish. Jobs left running hold resources others need.
- **Keys stay private.** Never paste a key into a chat, a repo, a shared doc or a screenshot.

If you get stuck

"no Nautilus context"	download the config at nrp.ai/config, rerun `nrp-mcp setup`
"forbidden" / not in namespace	sign in at nrp.ai once more; ask your namespace admin
`command not found`	open a new terminal, or use `~/.local/bin/nrp-mcp`
Missed the approval code	run `nrp-mcp approve` in your terminal
"Not ready yet: kubectl"	an older kubectl (Docker Desktop) is first on PATH; ask RC
Anything else	run `nrp-mcp doctor`; it names the next step
Laptop will not cooperate	use the NRP JupyterHub (only needs the sign-in): **jupyterhub-west.nrp-nautilus.io**

Prefer another assistant?

Run `nrp-mcp client` with `claude-desktop`, `claude-code`, `vscode`, `cursor`, `gemini` or `opencode`, then fully quit and reopen the app.

Gemini CLI needs Node.js 20+: `npm i -g @google/gemini-cli` and set `GEMINI_API_KEY`.

nrp-mcp
Install, Releases and the exercise in examples/
github.com/UCR-Research-Computing/nrp-mcp

Your NRP config
Sign in first, then Get Config; it lands in Downloads
nrp.ai/config

This sheet, online
Copyable commands, approval codes, other assistants; links to the full Nautilus guide
RC Knowledge Base, KB031